



Business Insights Report

Sales Performance Dashboard

Name: Prisha Kamatkar

Course: TYBCA

Internship Task: Task 1 – Sales Performance Dashboard

Date: 18 August 2026



Introduction

This report presents a detailed analysis of sales performance using data visualization and KPI tracking.

The objective is to identify key business insights across revenue, orders, product categories, payment methods, and regional performance.

Goals of the Report

- Provide an **Executive Overview** of overall business performance.
- Deliver a **Detailed Analysis** of product categories, payment preferences, and state-wise revenue.
- Highlight **Business Insights** that can guide decision-making and operational improvements.

Tools & Methods

- **Python (Jupyter Notebook):** Data cleaning, preprocessing, KPI calculations, and chart generation.
- **Power BI/Tableau:** Interactive dashboards for executive and detailed analysis.
- **GitHub:** Repository hosting and documentation.

```
In [ ]: # Step 1: Load the dataset first
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv("data1.csv") # replace with your file path

# Step 2: Inspect the raw columns
print(df.columns)

# Step 3: Standardize column names
df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_")
```

```
# Step 4: Check the updated column names
print(df.columns)
```

```
print("Missing values per column:\n", df.isnull().sum()) print("Duplicate rows:",
df.duplicated().sum())
```

```
In [3]: print(df.columns.tolist())
```

```
['order_id', 'date', 'customer_id', 'product_category', 'product_name', 'quantity',
'unit_price_inr', 'payment_method', 'delivery_status', 'review_rating', 'review_text',
'state', 'country', 'revenue=_quantity*_unitprice']
```

```
In [4]: # Create revenue column (if not already created)
```

```
df['revenue'] = df['quantity'] * df['unit_price_inr']
```

```
# Inspect key columns with correct names
```

```
print(df[['date', 'product_name', 'state', 'customer_id', 'revenue']].head())
```

	date	product_name	state	customer_id	revenue
0	25-01-2025	Cookware Set	Sikkim	CUST2796	51148.82
1	28-08-2025	Hair Dryer	Telangana	CUST9669	19361.41
2	27-02-2025	Tablet	Nagaland	CUST5808	115428.66
3	24-02-2025	Headphones	Assam	CUST5889	190728.60
4	15-06-2025	Saree	Odisha	CUST9005	229704.90

```
In [5]: print(df.columns.tolist())
```

```
['order_id', 'date', 'customer_id', 'product_category', 'product_name', 'quantity',
'unit_price_inr', 'payment_method', 'delivery_status', 'review_rating', 'review_text',
'state', 'country', 'revenue=_quantity*_unitprice', 'revenue']
```

```
In [6]: df = df.drop(columns=['revenue=_quantity*_unitprice'], errors='ignore')
```

```
In [7]: df['revenue'] = df['quantity'] * df['unit_price_inr']
```

```
In [8]: print(df[['date', 'product_name', 'state', 'customer_id', 'revenue']].head())
```

	date	product_name	state	customer_id	revenue
0	25-01-2025	Cookware Set	Sikkim	CUST2796	51148.82
1	28-08-2025	Hair Dryer	Telangana	CUST9669	19361.41
2	27-02-2025	Tablet	Nagaland	CUST5808	115428.66
3	24-02-2025	Headphones	Assam	CUST5889	190728.60
4	15-06-2025	Saree	Odisha	CUST9005	229704.90

```
In [9]: # Convert date column into proper datetime format
```

```
df['date'] = pd.to_datetime(df['date'], format='%d-%m-%Y')
```

```
# Create a Year-Month column
```

```
df['year_month'] = df['date'].dt.to_period('M')
```

```
# Calculate monthly revenue
```

```
monthly_revenue = df.groupby('year_month')['revenue'].sum()
```

```
# Display monthly revenue
```

```
print(monthly_revenue)
```

```

year_month
2025-01    92051785.03
2025-02    84995760.62
2025-03    93064349.39
2025-04    91388990.14
2025-05    97195848.48
2025-06    89730685.72
2025-07    95176904.42
2025-08    97576563.09
2025-09    91225555.74
2025-10    93478344.05
2025-11    94809186.29
2025-12    97467830.53
Freq: M, Name: revenue, dtype: float64

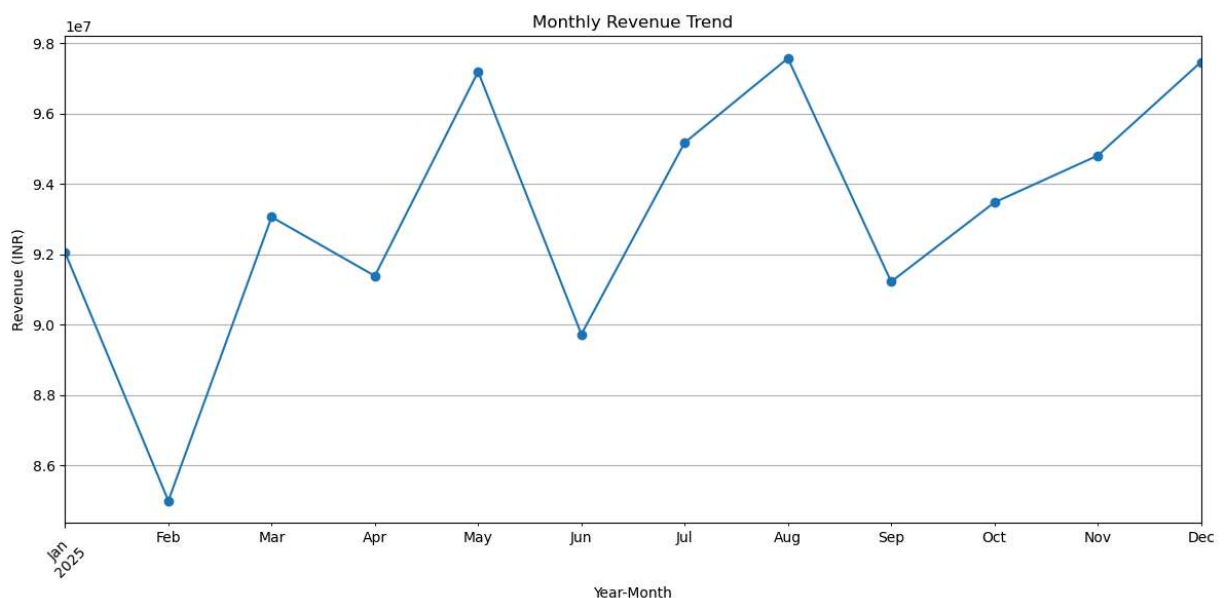
```

```

In [10]: # Plot monthly revenue trend
monthly_revenue.plot(
    kind='line',
    figsize=(12, 6),
    marker='o',
    title='Monthly Revenue Trend'
)

plt.xlabel('Year-Month')
plt.ylabel('Revenue (INR)')
plt.xticks(rotation=45)
plt.grid(True)
plt.tight_layout()
plt.show()

```



```

In [11]: # Calculate revenue by state
state_sales = (
    df.groupby('state')['revenue']
    .sum()
    .sort_values(ascending=False)
)

```

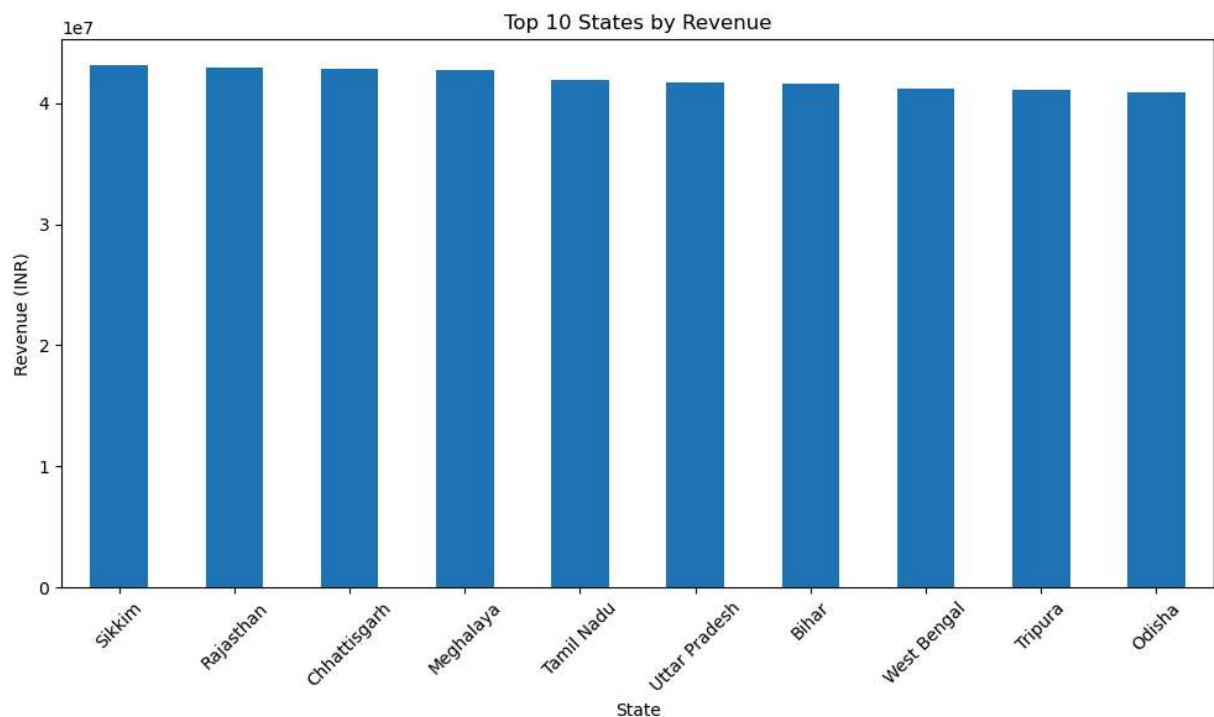
```
# Select top 10 states
top_10_states = state_sales.head(10)

print(top_10_states)
```

```
state
Sikkim      43113469.51
Rajasthan   42906175.08
Chhattisgarh 42857545.27
Meghalaya   42773152.96
Tamil Nadu  41967968.99
Uttar Pradesh 41690917.07
Bihar       41669240.44
West Bengal 41195932.45
Tripura     41103376.81
Odisha      40924381.38
Name: revenue, dtype: float64
```

```
In [12]: # Plot Top 10 States
top_10_states.plot(
    kind='bar',
    figsize=(10, 6),
    title='Top 10 States by Revenue'
)

plt.xlabel('State')
plt.ylabel('Revenue (INR)')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



```
In [13]: # Calculate revenue by product
product_sales = (
    df.groupby('product_name')['revenue']
```

```

    .sum()
    .sort_values(ascending=False)
)

# Select top 10 products
top_10_products = product_sales.head(10)

print(top_10_products)

```

```

product_name
Lipstick      48159020.15
Children's Book 48145664.52
Headphones    48044405.75
Hair Dryer    47428844.81
Perfume       46284732.62
Laptop        46173868.15
Tablet        46170022.15
Sneakers      46057295.93
Air Fryer     45919707.91
Jeans         45232819.47
Name: revenue, dtype: float64

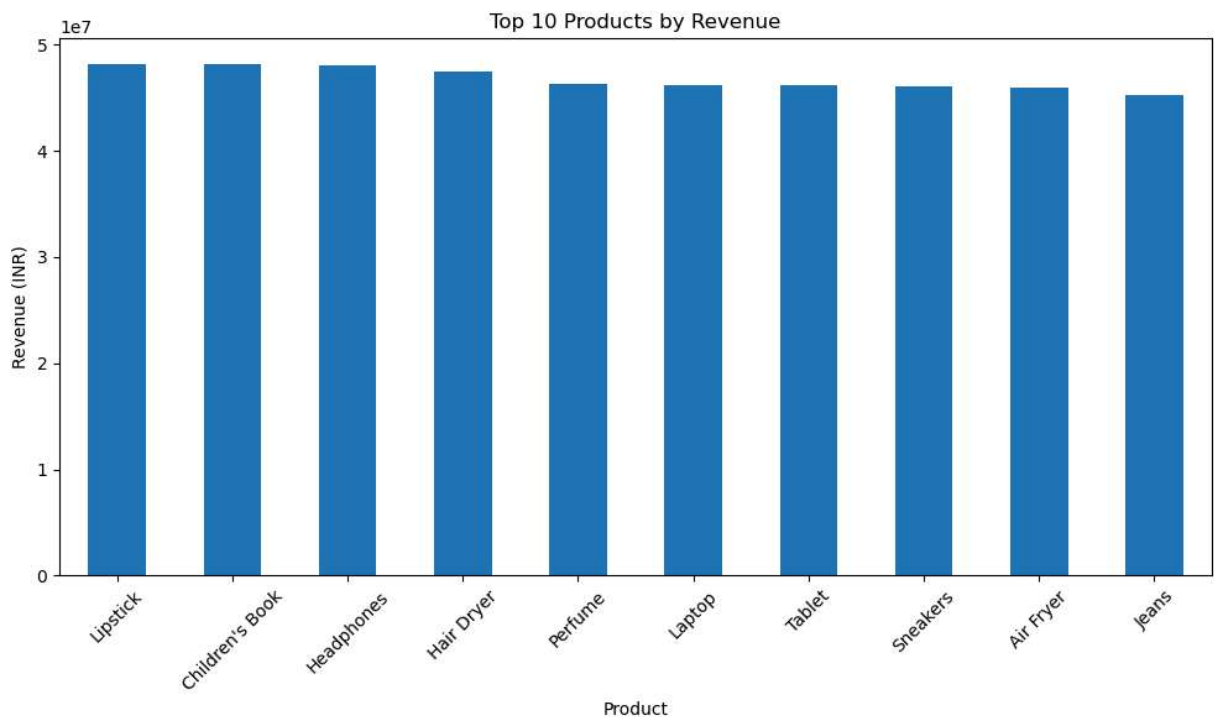
```

```

In [14]: # Plot Top 10 Products by Revenue
top_10_products.plot(
    kind='bar',
    figsize=(10, 6),
    title='Top 10 Products by Revenue'
)

plt.xlabel('Product')
plt.ylabel('Revenue (INR)')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```
In [15]: # Save the final cleaned dataset
df.to_csv("final_sales_data.csv", index=False)

print("Final dataset saved successfully!")
```

Final dataset saved successfully!



Business Insights Summary

- **Revenue Performance:** The business generated a total revenue of ₹1.12bn from 15K orders and 45K items sold.
- **Customer Spending:** The average order value is ₹74.54K, reflecting strong purchasing behavior.
- **Top Categories:** Beauty and Electronics are the leading contributors to revenue, showing consistent demand.
- **Payment Preferences:** Digital payments (Credit Card and UPI) together account for nearly half of total revenue, highlighting customer trust in online transactions.
- **Regional Trends:** States such as Rajasthan and Sikkim lead in revenue, while other regions show potential for growth.
- **Delivery Status:** Most orders are delivered successfully, but pending and returned orders (~33%) indicate areas for operational improvement.

Conclusion: Overall, the business is performing well with strong product categories and digital adoption. Future focus should be on reducing pending/returned orders and expanding in underperforming states to maximize growth opportunities.